

Classroom Problem-Solving Activity*Exception Handling: The Ticket Booth Validator*

Course topic	Exception Handling in Java
Concepts practised	try / catch / finally, multiple catch blocks, built-in exceptions, writing and throwing a custom checked exception
Format	Group work, in-class coding activity
Duration	35–40 minutes

1. Problem Statement

A cinema box office program asks the staff member to enter a customer's age (as text) and a seat number before selling a ticket. Three things can go wrong: the age might not be a number at all, the age might be unrealistic (below 0 or above 120), or the seat number might not exist in the seat list. Each of these is a different kind of problem and should be handled separately.

```
Enter customer age: abc
Error: that is not a valid number.
```

```
Enter customer age: -5
Error: age must be between 0 and 120.
```

```
Enter customer age: 34
Enter seat number (0-4): 9
Error: that seat number does not exist.
```

```
Enter customer age: 34
Enter seat number (0-4): 2
Ticket approved for age 34, seat A3.
Finished checking this customer.
```

2. What You Need to Build

1. A custom exception class `InvalidAgeException` that extends `Exception`, with a constructor that accepts a message and passes it to the superclass constructor.
2. A method `checkAge(String input)` that: (a) converts input to an int, (b) throws `InvalidAgeException` if the age is below 0 or above 120, and (c) prints a confirmation message if the age is valid.
3. A fixed array of five seat labels, e.g. `{"A1", "A2", "A3", "A4", "A5"}`.
4. A method `bookSeat(int seatNumber)` that looks up `seats[seatNumber]` and prints the seat label. Do not add your own bounds check here — let the array itself throw the exception.
5. A main method that calls `checkAge(...)` and `bookSeat(...)` inside one try block, with three separate catch blocks: `NumberFormatException`, `InvalidAgeException`, and `ArrayIndexOutOfBoundsException`.
6. Each catch block should print a clear, different message explaining what went wrong.
7. Add a finally block after the catch blocks that always prints "Finished checking this customer.", whether or not an exception occurred.
8. Test your program with: a non-numeric age, an age outside 0–120, a seat number outside the array's range, and a fully valid entry.